Variables of the Five Coded PTA Files

19 October 2026 · August Liu. The coding rules are in the workflow document; results are in the result report.

Files

File	Agreement	Rows
`ASEAN_JPN_coded_2026_10_19_v2.dta`	ASEAN–Japan	115,764
`BRN_JPN_coded_2026_10_19_v2.dta`	Brunei–Japan	83,491
`AUS_USA_coded_2026_10_19_v2.dta`	Australia–US	333,324
`CAFTA_coded_2026_10_19_v2.dta`	CAFTA (Central America–Dominican Republic–US)	321,967
`BHR_USA_coded_2026_10_19_v2.dta`	Bahrain–US	244,786

One row is one (restricted input, output) pair from the agreement's product-specific rules of origin, in the order of the original files. All codes are HS2002. The five files have the same 29 variables with the same storage types, and every variable has a Stata label, so they can be combined with `append`:

```
use ASEAN_JPN_coded_2026_10_19_v2, clear
foreach p in BRN_JPN AUS_USA CAFTA BHR_USA {
    append using `p'_coded_2026_10_19_v2
}
```

A pair has the same DV, uncertain, route and uncertain_reason in every agreement where it appears.

Variables

Variable	Type	Contents
`output_hs6`	str6	Output, HS2002 6-digit code (the original output_str)
`output_description`	strL	Output: HS2002 heading text | subheading text
`input_hs6`	str6	Restricted input, HS2002 6-digit code (the original input_str)
`input_description`	strL	Input: HS2002 heading text | subheading text
`VA_Alternative`	byte	From the original agreement file, unchanged
`VA_Complement`	byte	From the original agreement file, unchanged
`pta`	str	Agreement: ASEAN_JPN, BRN_JPN, AUS_USA, CAFTA, BHR_USA
`record_no`	long	Row number in the original file
`pair_id`	str	Unique row id: pta|input>output
`hs_vintage`	str	HS vintage of the codes (HS2002 for all five agreements)
`DV`	byte	**Main variable.** 1 = the input can be used to make the output; 0 = no link
`DV_strict`	byte	Conservative version: DV × (1 − uncertain), only certain links
`DV_excl_remelt`	byte	Robustness version: as DV, but direct remelting (route = recycle_direct, e.g. steel rod → steel ingots) = 0
`DV_max`	byte	Same as DV (kept for older code)

uncertain	byte	1 = kept at DV = 1 but doubtful; the reason is in uncertain_reason
uncertain_reason	str	Why uncertain = 1 (list below); empty when uncertain = 0
locked	byte	1 = input and output are the same good (workflow step 2), e.g. tea in bulk → tea in packets
step	double	Workflow step that set DV = 0: 3 unrelated goods, 4 parallel finished goods, 5 reverse direction, 6 no route; 2 = same good (locked); missing when DV = 1 through a route
route	str	Why the pair is 1, or none (list below)
source	str	How the code was obtained (list below)
policy	str	Convention of 19 October 2026 that changed the row, if any (list below)
ml_prob	double	Model probability that DV = 1 (information only); missing for ASEAN_JPN
ml_supported	double	1 = the model had enough similar coded pairs (more than 30 in the chapter pair) to be used; missing for ASEAN_JPN
concord_flag	byte	1 = a code is not a valid HS2002 subheading (290400 in ASEAN_JPN, 640410 in AUS_USA; 13 rows)
dv_ai1, dv_ai2	double	Code of AI coder 1 (Claude) and AI coder 2 (Codex) where they coded the pair; otherwise missing
why_ai1, why_ai2	strL	The two AI coders' one-sentence reasons
master_id	str	Id of the input–output pair (M|input>output), the same in every agreement

Values of route

Value	Meaning
same_good	Same good, or the same good further processed or packed (locked, step 2)
forward	The input is an earlier stage of the output (e.g. yarn → fabric of the same fibre)
ingredient	The input goes into the output as an ingredient or constituent
component	The input is a part of the output (e.g. nibs → pens, trims → garments)
mixture	The input is one component of a mixture or blend
sibling_intermediate	Two intermediate goods of the same material at the same stage
recycle_waste	Through a waste or scrap good
recycle_direct	Direct remelting with no scrap code in between (0 in DV_excl_remelt)
feed	The input feeds a farmed animal (e.g. feed fish → farmed salmon)
reconstitute	The output is the input reconstituted (e.g. dried → liquid egg albumin)
ml_forward, ml_ingredient, ml_component, ml_recycle, ml_sibling	Textile pair kept at 1 by the model and confirmed by the textile rules; the route predicted by the model
unresolved	Open case in the ASEAN–Japan labels (DV = 1, uncertain = 1)
none	No link (DV = 0)

Values of source

Value	Meaning
`pta1_label`	ASEAN–Japan labels: coded by hand and revised under the same conventions (ASEAN_JPN only)
`exact_copy`	Copied from the ASEAN–Japan labels (same HS6 pair)
`rule`	Input and output have the same code
`rule_textile`	Explicit textile rules (chapters 50–63)
`ai_units`, `ai_units_ml_reroute`, `ai_sample`	The two AI coders (Claude and Codex): heading-pair units, pairs moved from the model bands to the coders, and the audit sample
`ml`	Model confidence band, textile pairs confirmed by the textile rules
`conflict_rule`, `conflict_ml`, `conflict_unresolved`	Textile pairs where the model and the textile rules disagreed. conflict_rule: the rule was right in the audited sample, so the rule's code; conflict_ml: the model was right, so the model's code; conflict_unresolved: no clear winner, so the rule's code where it finds a route (policy P_CONF)

Values of uncertain_reason

Value	Meaning
minor constituent of a garment or made-up: other fibre or construction	Yarn or fabric of a fibre other than cotton or man-made fibres (wool, silk, flax, ramie, coarse hair, other), or of the same fibre but another construction (woven ↔ knitted), in a garment or made-up article
minor constituent of a yarn or fabric: other fibre	Fibre, yarn or waste of a fibre other than cotton or man-made fibres as a minor blend component of a yarn or fabric
AI coders, or the model and the textile rules, disagree	One AI coder found a production route and the other did not (or the model and the textile rules disagreed, and no convention settles the pair)
AI coders disagree: sibling or backward	One coder saw two intermediate siblings, the other a reverse (backward) pair
open case in the ASEAN-Japan labels	Left open in the original ASEAN–Japan hand coding (route = unresolved)
possible feed (rock lobster)	Crustaceans or crustacean meal → rock lobster: a possible but not established feed use

Values of policy

Conventions of 19 October 2026 (see the summary table of the workflow document). The field is empty when no convention changed the row.

Value	Convention
`P_Q1_1`	Intermediate goods of the same material at the same stage = 1 (pairs on which the AI coders disagreed)
`P_Q1_2`	Different synthetic fibres count as one fibre
`P_Q2_1`	Any chemical (chapters 28–38) → wastes of heading 3825 = 1
`P_Q2_2`	Any mineral of chapter 25 → cement (2523) = 1

P_Q4	Direct remelting = 1 in DV (route recycle_direct)
P_CONF	Model and textile rules disagreed and the rule finds a route: the rule’s code
P_UNC	Kept at 1 by the textile rules or the model, marked uncertain as a minor constituent of another fibre
P_AUD_A	Minor fibre, yarn or waste of another fibre (not cotton or man-made) in a yarn or fabric = 1, uncertain
P_AUD_B	Intermediate goods of the same material coded 0 by both AI coders = 1
P_AUD_C	Sewing thread ↔ yarn and twine of the same fibre = 1
P_OPT2	Cotton or man-made fibre, yarn, fabric or waste as a minor constituent = 1 (certain)
P_FISH	Fish and other seafood of different species = 0 (step 3), unless the input could feed the output